OOPS ASSIGNMENT 1

NAME: Puneet Singh

ROLL NUMBER: 1024150037

BATCH: 2O11

1. Write a program (WAP) to display "Hello World" on the console display.

main.cpp

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "Hello World.\n";
    cout << "Puneet Singh, 1024150037";

    return 0;
}
```

Output

```
Hello World.
Puneet Singh, 1024150037

=== Code Execution Successful ===
```

2. WAP to read two values from the keyboard and display the larger value on the screen.

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;

    if (a > b)
        cout << "Larger value: " << a << endl;
    else
        cout << "Larger value: " << b << endl;

    cout << "Puneet Singh (1024150037)";
    return 0;
}
```

Output

```
Enter two numbers: 5
7
Larger value: 7
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

3. WAP to read a, b, c and calculate x = a/b – c

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    float a, b, c, x;
    cout << "Enter a, b, c: ";
    cin >> a >> b >> c;

    x = a / b - c;
    cout << "Value of x = " << x << endl;
    cout << "Puneet Singh (1024150037)";

    return 0;
}
```

Output

```
Enter a, b, c: 5
10
15
Value of x = -14.5
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

4. WAP to implement the following control characters: - ‘\n’ is for new line, or you can use endl – cout<<endl<<“message”; -
   ‘\t’ is for tab
   ‘\a’ is an alarm sound
   ‘\r’ is a carriage return to go to the beginning of the current line

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Hehe\nWohoo" << endl;
    cout << "Hello\tWorld" << endl;
    cout << "1024150037\a" << endl;
    cout << "OOPS\\ASSIGNMENT" << endl;

    cout << "Puneet Singh (1024150037)";
    return 0;
}


// cout << "Puneet Singh (1024150037)";
```

Output

```
Hehe
Wohoo
Hello	World
1024150037
OOPS\ASSIGNMENT
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

5. Implement namespace in a program to illustrate the use of the same name variables and functions in different sections/libraries of the code.

main.cpp

```cpp
#include <iostream>
using namespace std;

namespace one {
    int a = 10;
}

namespace two {
    int a = 20;
}

int main() {
    cout << "Value from namespace one: " << one::a << endl;
    cout << "Value from namespace two: " << two::a << endl;

    cout << "Puneet Singh (1024150037)";
    return 0;
}
```

Output

```
Value from namespace one: 10
Value from namespace two: 20
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

6. Write a program to define a structure Student that contains 3 variables and 2 functions: - Variables: (a) name, (b) roll, (c) marks - Functions: (1) setStudentData(), (2) getStudentData(). Use loops to input and output data for 3 student object entries.

```cpp
struct Student {
    string name;
    int roll;
    int marks;

    void setStudentData() {
        cin >> name >> roll >> marks;
    }

    void getStudentData() {
        cout << name << " " << roll << " " << marks << endl;
    }
};

int main() {
    Student s[3];

    cout << "Enter name roll marks of 3 students:\n";
    for (int i = 0; i < 3; i++) {
        s[i].setStudentData();
    }
```

```
Enter name roll marks of 3 students:
Puneet 15 100
Ayush 13 99
Soham 35 98
Student details:
Puneet 15 100
Ayush 13 99
Soham 35 98
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

7. In the previous program, make the data private, and functions public.

main.cpp

```cpp
class Student {
private:
    string name;
    int roll;
    int marks;

public:
    void setData() {
        cin >> name >> roll >> marks;
    }

    void showData() {
        cout << name << " " << roll << " " << marks << endl;
    }
};

int main() {
    Student s[3];

    cout << "Enter details:\n";
    for (int i = 0; i < 3; i++) {
```

Output

```
Enter details:
P 1 100
A 13 99
S 35 98
Student details:
P 1 100
A 13 99
S 35 98
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

8.

8. Convert program #4 from struct to class.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Student {
public:
    string name;
    int roll;
    int marks;

    void setData() {
        cin >> name >> roll >> marks;
    }

    void showData() {
        cout << name << " " << roll << " " << marks << endl;
    }
};

int main() {
    Student s[3];

    cout << "Enter details:\n";
    for (int i = 0; i < 3; i++) {
        s[i].setData();
    }

    cout << "Student details:\n";
    for (int i = 0; i < 3; i++) {
        s[i].showData();
    }

    cout << "Puneet Singh (1024150037)";
    return 0;
}
```

Output

```
Enter details:
A 1 85
C 5 85
C 22 95
Student details:
A 1 85
C 5 85
C 22 95
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

9. Discuss the differences between structures in C and structures in C++ during the lab with other students or your instructor.

ANSWER:

- In C, a structure can store only data.
- In C++, a structure can store data and functions.
- Access specifiers are not available in C.
- Access specifiers (public/private) are available in C++.
- C++ supports object-oriented programming.

10. WAP to read an integer value from the keyboard and display on screen “WELL DONE”

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int x, i;
    cout << "Enter an integer: ";
    cin >> x;

    for (i = 1; i <= x; i++) {
        cout << "WELL DONE" << endl;
    }

    cout << "Puneet Singh (1024150037)";
    return 0;
}
```

Output

```
Enter an integer: 4
WELL DONE
WELL DONE
WELL DONE
WELL DONE
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

11. WAP to use assignment operators += and - =

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int a = 10;

    a += 5;
    cout << "After += : " << a << endl;

    a -= 3;
    cout << "After -= : " << a << endl;

    cout << "Puneet Singh (1024150037)";
}
```

Output

```
After += : 15
After -= : 12
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

12. WAP to swap two numbers using the bitwise operator.

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;

    a = a ^ b;
    b = a ^ b;
    a = a ^ b;

    cout << "After swapping: " << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;

    cout << "Puneet Singh (1024150037)";
    return 0;
}
```

Output

```
Enter two numbers: 5
10
After swapping:
a = 10
b = 5
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

13. Library fine calculation program (simple if-else).

main.cpp

Share Run

```cpp
using namespace std;

int main() {
    int days;
    cout << "Enter number of late days: ";
    cin >> days;

    if (days <= 5) {
        cout << "Fine = " << days * 0.50 << " rupees" << endl;
    }
    else if (days <= 10) {
        cout << "Fine = " << days * 1 << " rupees" << endl;
    }
    else if (days <= 30) {
        cout << "Fine = " << days * 5 << " rupees" << endl;
    }
    else {
        cout << "Membership Cancelled" << endl;
    }

    cout << "Puneet Singh (1024150037)";
    return 0;
```

Output

```
Enter number of late days: 15
Fine = 75 rupees
Puneet Singh (1024150037)

=== Code Execution Successful ===
```

ASSIGNMENT-2

Name- Puneet Singh

Roll No: 1024150037

Q1.

main.cpp

```cpp
#include <iostream>
using namespace std;
struct Student {
    string name;
    int rollNo;
    string degree;
    string hostel;
    float cgpa;
    void addDetails() {
        cout << "Enter Name: ";
        cin >> name;
        cout << "Enter Roll No: ";
        cin >> rollNo;
        cout << "Enter Degree: ";
        cin >> degree;
        cout << "Enter Hostel: ";
        cin >> hostel;
        cout << "Enter CGPA: ";
        cin >> cgpa;
    }
    void updateCGPA(float newCGPA) {
        cgpa = newCGPA;
    }
    void updateHostel(string newHostel) {
        hostel = newHostel;
    }

    void displayDetails() {
        cout << "\nName: " << name
             << "\nRoll No: " << rollNo
             << "\nDegree: " << degree
             << "\nHostel: " << hostel
             << "\nCGPA: " << cgpa << endl;
    }
};
int main() {
    cout<<"Puneet Singh, 1024150037"<<endl;
    Student s;
```

Output

```
Puneet Singh, 1024150037
Enter Name: Soham
Enter Roll No: 35
Enter Degree: ENC
Enter Hostel: D
Enter CGPA: 9

Name: Soham
Roll No: 35
Degree: ENC
Hostel: Neeram Hall
CGPA: 9


=== Code Execution Successful ===
```

Q2.

main.cpp

```cpp
#include <iostream>
using namespace std;
class Student {
private:
    string name;
    int rollNo;
    float cgpa;

    void updateCGPA(float c) {
        cgpa = c;
    }
public:
    void addDetails() {
        cout << "Enter Name, Roll No, CGPA: ";
        cin >> name >> rollNo >> cgpa;
    }
    void modifyCGPA(float c) {
        updateCGPA(c);
    }
    void display() {
        cout << name << " " << rollNo << " " << cgpa << endl;
    }
};
int main() {
    cout<<"Puneet Singh, 1024150037"<<endl;
    Student s;
    s.addDetails();
    s.modifyCGPA(8.5);
    s.display();
}
```

Output

```
Puneet Singh, 1024150037
Enter Name, Roll No, CGPA: Puneet
37
9
Puneet 37 8.5


=== Code Execution Successful ===
```

Q3.

main.cpp

```cpp
#include <iostream>
using namespace std;
class Demo {
private:
    void secret() {
        cout << "Private function called by Puneet Singh, 1024150037\n";
    }
public:
    void show() {
        secret();
    }
};
int main() {

    Demo d;
    d.show();
}
```

Output

```
Private function called by Puneet Singh, 1024150037


=== Code Execution Successful ===
```

Q4.

main.cpp

```cpp
#include <iostream>
using namespace std;
class Rectangle {
int width, height;
public:
    void getdata() {
        cout << "Enter width: ";
        cin >> width;
        cout << "Enter height: ";
        cin >> height;
    }
    void calculatearea() {
        cout << "Area = " << width * height << endl;
    }
};
int main() {
    Rectangle r;
    r.getdata();
    r.calculatearea();
    cout<<"Puneet Singh, 1024150037";
}
```

Output

```
Enter width: 5
Enter height: 10
Area = 50
Puneet Singh, 1024150037

=== Code Execution Successful ===
```

Q5.

main.cpp

```cpp
#include <iostream>
using namespace std;
class Complex {
    float real, imag;
public:
    void setComplex(float r, float i) {
        real = r;
        imag = i;
    }
    void displayComplex() {
        cout << real << " + " << imag << "i\n";
    }
    Complex add(Complex c) {
        Complex temp;
        temp.real = real + c.real;
        temp.imag = imag + c.imag;
        return temp;
    }
};
int main() {
    cout<<"Puneet Singh, 10124150037"<<endl;
    Complex c1, c2, sum;
    c1.setComplex(2, 5);
    c2.setComplex(4, 6);
    sum = c1.add(c2);
    sum.displayComplex();
}
```

Output

```
Puneet Singh, 10124150037
6 + 11i


=== Code Execution Successful ===
```

Q6.a)

main.cpp

```cpp
#include <iostream>
using namespace std;
class Test {
public:
    void show();
};
void Test::show() {
    cout << "Function defined outside class" << endl;
}
int main() {
     cout<<"Puneet Singh, 1024150037"<<endl;
    Test t;
    t.show();
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Function defined outside class


=== Code Execution Successful ===
```

Q6.b)

main.cpp

```cpp
#include <iostream>
 using namespace std;
 int x = 10;
int main() {
    int x = 20;
     cout<<"Puneet Singh, 1024150037"<<endl;
    cout << "Local x: " << x << endl;
    cout << "Global x: " << ::x << endl;
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Local x: 20
Global x: 10


=== Code Execution Successful ===
```

Q6.c)

main.cpp

```cpp
#include <iostream>
 using namespace std;
 class variable {
 public:
    static int x;
 };
 int variable::x = 50;
 int main() {
     cout<<"Puneet Singh, 1024150037"<<endl;
    cout << variable::x << endl;
    return 0;
 }
```

Output

```
Puneet Singh, 1024150037
50


=== Code Execution Successful ===
```

Q6.d)

main.cpp

```cpp
#include <iostream>
int main() {
    std:: cout<<"Puneet Singh, 1024150037";
    return 0;
}
```

Output

```
Puneet Singh, 1024150037

=== Code Execution Successful ===
```

Q7.

main.cpp

```cpp
#include <iostream>
using namespace std;
namespace A {
    int x = 10;
    void show() {
        cout << "Namespace A\n";
    }
}
namespace B {
    int x = 20;
    void show() {
        cout << "Namespace B\n";
    }
}
int main() {
    cout<<"Puneet Singh, 1024150037"<<endl;
    cout << A::x << endl;
    cout << B::x << endl;
    A::show();
    B::show();
}
```

Share Run

Output

```
Puneet Singh, 1024150037
10
20
Namespace A
Namespace B


=== Code Execution Successful ===
```

OOPS ASSIGNMENT – 3

Name: Puneet Singh

Roll No. :1024150037

2O11

Lab Assignment 3

1. Write a program to implement
   (a) pointer to an object
   (b) this pointer.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Demo {
    int x;
public:
    void set(int x) {
        this->x = x;
    }
    void show() {
        cout << x << endl;
    }
};

int main() {
    Demo obj;
    Demo *ptr;
    ptr = &obj;
    obj.set(10);
    ptr->show();
    cout << "Puneet Singh, 1024150037";
    return 0;
}
```

Output

```
10
Puneet Singh, 1024150037

=== Code Execution Successful ===
```

Practice both . (dot operator) and -> (arrow operator).

2. Write a program to swap private values of two classes using a friend function.

main.cpp

```cpp
#include <iostream>
using namespace std;

class B;

class A {
    int x;
public:
    A(int a) { x = a; }
    friend void swap(A&, B&);
};

class B {
    int y;
public:
    B(int b) { y = b; }
    friend void swap(A&, B&);
};

void swap(A& a, B& b) {
    int t = a.x;
    a.x = b.y;
    b.y = t;
    cout << a.x << endl;
    cout << b.y << endl;
}

int main() {
    A a(10);
    B b(20);
    swap(a, b);
    cout << "Puneet Singh, 1024150037";
    return 0;
}
```

Output

```
20
10
Puneet Singh, 1024150037

=== Code Execution Successful ===
```

3. Write a program to add data objects of two different classes using friend functions.

main.cpp

```cpp
#include <iostream>
using namespace std;

class B;

class A {
    int x;
public:
    A(int a) { x = a; }
    friend void add(A, B);
};

class B {
    int y;
public:
    B(int b) { y = b; }
    friend void add(A, B);
};

void add(A a, B b) {
    cout << a.x + b.y << endl;
}

int main() {
    A a(5);
    B b(7);
    add(a, b);
    cout << "Puneet Singh, 1024150037";
    return 0;
}
```

Output

```
12
Puneet Singh, 1024150037

=== Code Execution Successful ===
```

4. Write a program to demonstrate the working of friend class.

main.cpp

```cpp
#include <iostream>
using namespace std;

class B;

class A {
    int x;
public:
    A() { x = 50; }
    friend class B;
};

class B {
public:
    void show(A a) {
        cout << a.x << endl;
    }
};

int main() {
    A a;
    B b;
    b.show(a);
    cout << "Puneet Singh, 1024150037";
    return 0;
}
```

Output

```
50
Puneet Singh, 1024150037

=== Code Execution Successful ===
```

5. Write a program using Array of Objects to display area of multiple rectangles.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Rectangle {
    int l, b;
public:
    void set() {
        cin >> l >> b;
    }
    void area() {
        cout << l * b << endl;
    }
};

int main() {
    Rectangle r[3];
    for (int i = 0; i < 3; i++)
        r[i].set();

    for (int i = 0; i < 3; i++)
        r[i].area();

    cout << "Puneet Singh, 1024150037";
    return 0;
}
```

input

```
23
56
45
45
585
1288
2025
Puneet Singh, 1024150037

...Program finished with exit code 0
Press ENTER to exit console.
```

6. Write a program to define function cube() as inline for calculating cube of a number.

Run Debug Stop Share Save { } Beautify

main.cpp

```cpp
#include <iostream>
using namespace std;

inline int cube(int n) {
    return n * n * n;
}

int main() {
    int n;
    cin >> n;
    cout << cube(n) << endl;
    cout << "Puneet Singh, 1024150037";
    return 0;
}
```

input

```
15
3375
Puneet Singh, 1024150037

...Program finished with exit code 0
Press ENTER to exit console.
```

7. Write a program to pass an object as an argument and return the object from a function.
   a. Use pass-by-value.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Sample {
public:
    int x;
};

Sample change(Sample m) {
    m.x = m.x + 10;
    return m;
}

int main() {
    Sample a, b;
    a.x = 20;
    b = change(a);
    cout << a.x << endl;
    cout << b.x << endl;
    cout << "Puneet Singh, 1024150037";
    return 0;
}
```

Output

```
20
30
Puneet Singh, 1024150037

=== Code Execution Successful ===
```

b. Use pass-by-address

main.cpp

```cpp
#include <iostream>
using namespace std;

class Sample {
public:
    int x;
};

Sample change(Sample *m) {
    m->x = m->x + 10;
    return *m;
}

int main() {
    Sample a, b;
    a.x = 20;
    b = change(&a);
    cout << a.x << endl;
    cout << b.x << endl;
    cout << "Puneet Singh, 1024150037";
    return 0;
}
```

Output

```
30
30
Puneet Singh, 1024150037

=== Code Execution Successful ===
```